A Project Synopsis

on

“SMART PLANNER LMS”

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Submitted by

AYUSH CHAURASIYA

(B2255R10106007)

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Under the Guidance of

Dr. Akhilesh A. Waoo

Dean CS, AKS University, SATNA, MP

Department of Computer Science & Engineering
Faculty of Engineering and Technology

AKS University, Satna (M.P)

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ABSTRACT

The LMS Smart Planner is a web-based academic management and productivity system developed to simplify the daily activities of students and teachers. In many educational environments, students use different applications for assignments, notes, reminders, and study planning, which often creates confusion and reduces productivity. The main purpose of this project is to combine all important academic tools into a single platform that is simple, organized, and easy to use.

The system allows students to manage tasks, access study materials, track progress, and improve concentration using a built-in focus timer. Teachers can upload course materials, assign tasks, monitor student performance, and provide feedback directly through the platform. Administrators manage users and maintain the overall system efficiently.

The LMS Smart Planner is developed using React.js for the frontend, FastAPI for the backend, and MySQL for database management. The frontend provides a responsive and user-friendly interface, while the backend handles data processing and communication between the user interface and database. MySQL stores all important information such as user details, tasks, reports, and course materials securely.

The system is designed to improve organization, time management, and communication within the academic environment. Features such as task boards, progress tracking, and study timers help students stay focused and productive. The platform also reduces the need for multiple applications by providing all essential features in one place.

INTRODUCTION

In today's modern educational environment, technology plays an important role in improving the learning process and managing academic activities. Students and teachers depend heavily on digital platforms for assignments, study materials, communication, and progress tracking. However, most students still use multiple applications for different purposes such as task management, note-taking, reminders, and focus timers. Managing all these tools separately often creates confusion, wastes time, and reduces productivity.

The LMS Smart Planner is developed to solve these problems by providing a single platform that combines learning management and productivity tools into one system. The project is designed as a web-based application that helps students organize their academic activities more effectively while also helping teachers manage courses and monitor student progress. The system focuses on simplicity, organization, and ease of use so that users can easily access all important features from one place.

The LMS Smart Planner provides separate dashboards for students, teachers, and administrators. Students can manage assignments, create tasks, access course materials, and use the built-in focus timer to improve concentration during study sessions. Teachers can upload study resources, assign tasks, monitor student performance, and provide feedback directly through the platform. Administrators are responsible for managing users and maintaining the smooth functioning of the system.

One of the important features of the project is the task management system, which allows students to organize their assignments visually using categories such as To-Do, In Progress, and Completed. Another useful feature is the focus timer based on the Pomodoro technique, which helps students study in structured time intervals and maintain better concentration.

The project is developed using modern web technologies. React.js is used for the frontend to create a responsive and interactive user interface. FastAPI with Python is used for backend development to handle server-side operations and data processing. MySQL is used as the database for storing user information, tasks, reports, and course materials securely.

OBJECTIVES

The major objectives of the project are as follows:

- To create a single platform for managing academic activities
- To help students organize assignments and daily study tasks
- To improve time management using a built-in focus timer
- To provide easy access to study materials and course resources
- To simplify task tracking and progress monitoring
- To improve communication between students and teachers
- To reduce confusion caused by using multiple applications
- To provide a clean and user-friendly interface
- To allow teachers to assign tasks and monitor student performance
- To maintain secure storage of user and academic data

FEASIBILITY STUDY

Technical Feasibility

• Required technologies like React.js, FastAPI, Python, and MySQL are easily available

• The system can run on standard laptops and desktops

• Development tools are simple and widely supported

• The project can be maintained and updated easily

• Internet and browser support are commonly available

Economic Feasibility

• The project uses open-source technologies

• Development cost is low

• No expensive hardware is required

• Maintenance cost is affordable

• Reduces manual work and saves time

Operational Feasibility

• The system is simple and user-friendly

• Students and teachers can use it easily

• Improves academic management and organization

• Supports smooth communication between users

• Helps increase productivity and focus

Schedule Feasibility

• Project development followed a planned schedule

• Modules were completed step by step

• Testing was performed during development

• Resources and time were managed properly

• The project can be completed within the given timeline

MODULES OF PROJECT / PLANING OF WORK

Modules of the Project

Module Name	**Description**
User Module	Handles user registration, login, and role management for Students, Teachers, and Admins
Task Management Module	Allows students to create, update, and manage assignments and daily tasks
Focus Timer Module	Provides a study timer to improve concentration and productivity
Reports Module	Tracks student progress and allows teachers to provide feedback

Planning of Work

Phase	**Work Performed**
Requirement Analysis	Identified project goals and user requirements
System Design	Planned frontend interface and database structure
Development	Developed frontend, backend, and database modules
Testing	Tested all modules and fixed errors
Implementation	Executed the final system and prepared documentation

HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements

Components	**Minimum Requirement**
Processor	Intel i3 or higher / Equivalent AMD
RAM	4 GB minimum (8 GB recommended for VMs)
Hard Disk	20 GB free space
Monitor	1024×768 resolution or higher
Input Devices	Standard Keyboard and Mouse
Internet Connection	Stable broadband with SSH access

Software Requirements

Operating System (Windows/Linux/macOS)	Platform to run the application
Python 3.x	Backend programming language
FastAPI	Backend framework for server-side development
React.js	Frontend library for user interface
MySQL	Database management system
Visual Studio Code	Code editor for development
Google Chrome / Firefox / Edge	Web browser to access the system
Node.js and npm	Required for React frontend setup

PROJECT PLANNING

Week 1

During the first week, the project topic was selected and the main problems faced by students in academic management were identified. Basic research was performed to understand the need for a centralized academic planning system. Project objectives and requirements were also discussed during this phase.

• Selected project topic

• Identified academic management problems

• Collected initial project requirements

• Defined project objectives

Week 2

In the second week, the overall system structure and important features of the LMS Smart Planner were planned. Different modules required for the project were identified and organized properly.

• Planned project modules

• Designed system workflow

• Selected major project features

• Planned user roles and functionalities

Week 3

The third week focused on designing the frontend interface and user experience. Dashboard layouts, navigation flow, and page structure were planned to make the system simple and user-friendly.

• Designed frontend layout

• Planned dashboard structure

• Created navigation flow

• Designed user interface structure

Week 4

During the fourth week, database planning and backend connectivity were started. Database tables and relationships were designed for storing user data, tasks, courses, and reports.

• Designed database tables

• Planned entity relationships

• Started backend connectivity

• Configured MySQL database

Week 5

The fifth week focused on developing the main modules of the system, especially the User Module and Task Management Module.

• Developed login and registration system

• Implemented user role management

• Developed task creation and management features

• Connected frontend with backend APIs

Week 6

In the sixth week, additional modules such as the Focus Timer Module, Course Module, and Reports Module were developed and integrated into the system.

• Developed focus timer functionality

• Added course material management

• Implemented reports and feedback system

• Integrated multiple modules together

Week 7

The seventh week was mainly used for system testing and debugging. Different modules were tested individually and as a complete system to identify and fix errors.

• Performed login testing

• Tested task management features

• Verified database operations

• Performed security and performance testing

Week 8

The final week focused on completing the project implementation, improving performance, and preparing the project documentation and report.

• Fixed remaining system errors

• Improved overall system performance

• Integrated all project modules completely

• Prepared final documentation and report submission

FUTURE SCOPE

Future Enhancements

- Mobile application support
- AI-based study recommendations
- Smart task prioritization
- Notification and reminder system
- Advanced analytics and reports
- Online meeting and video lecture support
- Chat and communication features
- Cloud storage integration
- Dark mode and UI customization
- Multi-language support

REFERENCES

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• React.js Official Documentation – https://react.dev/

• FastAPI Official Documentation – https://fastapi.tiangolo.com/

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• W3Schools Web Development Tutorials – https://www.w3schools.com/